$\frac{3-\sqrt{17}}{2}$ by the Robust Thm!

Thus $\alpha(f_c, z_0) < \alpha_0$ and $N_{f_c}^k(z_0)$ converges to the unique root ζ_c of f_c in the open ball of radius ~~[illegible]~~$/\gamma(f, z_0)$ around z_0. Moreover $f(\zeta_c) = c + f(z_0)$ and $f'(\zeta_c)^{-1}$ exists by Lemma 2. ~~The proposition follows.~~ In particular, □

$$f_{z_0}^{-1\prime}(c) = \lim_{w \to c} \frac{\zeta_w - \zeta_c}{w - c} = \lim_{w \to c} \lim_{k \to \infty} \frac{N_{f_w}^k(z_0) - N_{f_c}^k(z_0)}{w - c} = \lim_{k \to \infty} \frac{\partial}{\partial w} N_{f_w}^k(z_0)\Big|_c$$

by uniform cgce, via Robust Thm.

8.3 Additional Comments and Bibliographical Remarks

Kantorovich pioneered a general modern treatment of Newton's method (see [Kantorovich and Akilov 1964]). The approach here is modeled after the paper by Smale [1986b] which emphasizes data at one point in contrast to Kantorovich hypotheses of estimates over a region.

Earlier work is [Traub and Woźniakowski 1979] and [Woźniakowski 1977]. Theorem 2 is proved in [Smale 1986b] with the constant $\alpha_0 = 0.130707$ and Kim [1988] has a one-dimensional version. The best constant for α_0 (equal to $\frac{1}{4}(13 - 3\sqrt{17}) \approx 0.157671$)) was subsequently found by Royden [1986] and Wang Xinghua with Han Danfu (see [Wang 1993]). In [Shub and Smale 1985, 1986] this last way is presented with the best constants for α_0. Curry [1989] and Chen [1994] have found generalizations.

The first version of *approximate zero* in the way it is used here is in [Smale 1981] with various developments in [Shub and Smale 1985, 1986]. Some applications and other developments of this α-theory were given in [Smale 1986a; Rheinboldt 1988; Renegar and Shub 1992; Ye 1994]. The approach to proving Theorems 3 and 4 is in [Shub and Smale 1994].

Dedieu [TAb] has a version of the separation theorem, Corollary 1. See also [Malajovich-Muñoz 1993].

This chapter covers only one approach to Newton's method, one which we find useful for complexity analysis. But there is a vast and fascinating literature on the subject of Newton's method which we have ignored here.

~~[illegible]~~

$$= \cdots = f'(\zeta_c)^{-1}. \quad (!)$$

9 Fundamental Theorem of Algebra: Complexity Aspects

This chapter is devoted to producing a complexity analysis of a "homotopy method" for finding approximately a zero of a polynomial. The homotopy method is realized as a sequence of applications of Newton's method. This algorithm is a prototype of the main methods of numerical analysis for solving a nonlinear system of equations. The complexity analysis here (and extensions in later chapters) gives a depth of understanding of these methods which is missing in the usual treatment based on convergence proofs. In this chapter we only consider the one-variable case. We use from the previous chapter only the easily proved Proposition 2. We also use a result from the theory of Schlicht functions due to Loewner.

9.1 The Fundamental Theorem of Algebra

The fundamental theorem of algebra asserts that every nonconstant polynomial with complex numbers as coefficients has a complex zero. In other words, for every nonconstant complex polynomial f, there is a point ζ in the complex plane with $f(\zeta) = 0$.

In this section we give a proof of it. The proof gives rise to a numerical algorithm to find an approximate zero and we analyze the algorithm from a complexity perspective, using the theorems we have proved about Newton's method in the last chapter. We do not attempt to get the best constants or most efficient algorithms (even so our algorithms are quite efficient) but rather to illustrate some of the principles of a complexity theory for numerical analysis.

The main tool is the inverse function theorem for one variable which is used in the following form. Let f be a complex polynomial and $z \in \mathbb{C}$ with $f'(z) \neq 0$. Then there is a $\delta > 0$ and a complex differentiable (i.e., complex analytic) map

$$f_z^{-1} : D_\delta(f(z)) \to \mathbb{C}$$

with $f_z^{-1}(f(z)) = z$ and $f(f_z^{-1}(w)) = w$ for all $w \in D_\delta(f(z))$. Here $D_\delta(f(z))$ is the set of all w such that $|w - f(z)| < \delta$.

Moreover, if z ranges over a closed and bounded set $K \subset \mathbb{C}$ where f' is never zero, then the corresponding $\delta > 0$ can be chosen independent of $z \in K$. The proof of this last fact can be obtained by taking a convergent subsequence as in usual compactness arguments.

A complex polynomial of *degree d* has an expression

$$f(z) = \sum_{i=0}^{d} a_i z^i, \quad a_d \neq 0, \quad a_i \in \mathbb{C} \text{ for all } i. \tag{9.1}$$

Theorem 1 (Fundamental Theorem of Algebra) *Let f be a complex polynomial of degree d, $d \geq 1$. Then there is $\zeta \in \mathbb{C}$ with $f(\zeta) = 0$.*

First we prove two lemmas about such polynomials.

Lemma 1 *The polynomial f has at most d roots.*

Proof. Suppose now that $f(\zeta) = 0$. Then by polynomial division we may write $f(z) = (z - \zeta)g(z)$ where g is a polynomial with degree $d - 1$. Now induction finishes the proof. □

Lemma 2 *Let f be as in* (9.1). *If $f(\zeta) = 0$, then*

$$|\zeta| < 2 \max_{1 \leq k \leq d} \left(\left| \frac{a_{d-k}}{a_d} \right|^{1/k}, 1 \right).$$

Proof. Let

$$b = \max_{1 \leq k \leq d} \left(\left| \frac{a_{d-k}}{a_d} \right|^{1/k}, 1 \right) \quad \text{and} \quad g(z) = \frac{1}{a_d b^d} f(bz).$$

Then

$$g(z) = \sum_{i=0}^{d} c_i z^i, \quad c_d = 1, \quad |c_i| \leq 1 \text{ all } i.$$

Thus

$$\left| \sum_{0}^{d-1} c_i z^i \right| < \sum_{k=0}^{d-1} |z|^k = \frac{|z|^d - 1}{|z| - 1}$$

and the last is less than $|z|^d$ for $|z| \geq 2$. Hence $g(z) \neq 0$ for $|z| \geq 2$ and $f(z) \neq 0$ for $|z|/b \geq 2$. □

We recall that a map $f : \mathbb{C}^n \to \mathbb{C}^m$ is said to be *proper* when the pre-image $f^{-1}(S)$ of every closed and bounded set $S \subset \mathbb{C}^m$ is also closed and bounded. An immediate consequence of the preceding lemma is the following result.

Corollary 1 *A nonconstant polynomial $f : \mathbb{C} \to \mathbb{C}$ is proper.* □

Definition 1 A *critical point* θ of a polynomial f is a point where the derivative f' vanishes. If f has degree $d > 0$, then f' is a polynomial of degree $(d - 1)$ and so has at most $(d - 1)$ zeros. Thus f has at most $(d - 1)$ critical points. A *critical value* of f is a point $f(\theta)$, where θ is some critical point. It follows that a polynomial of degree d has at most $(d - 1)$ critical values.

Toward the proof of Theorem 1, we may assume that 0 is not a critical value of f, since if $f'(\theta) = 0$ and $f(\theta) = 0$, θ is our desired solution.

Next we say that z_0 is a *good point* (of f) if the line segment

$$L = \{tf(z_0) \mid 0 \leq t \leq 1\} = L_{z_0} \tag{9.2}$$

contains no critical value of f.

Let us prove that there are good points. There are $z_1 \in \mathbb{C}$ that are not critical points of f. By the inverse function theorem, the image of a disk around z_1 contains a disk around $f(z_1)$. The image disk contains infinitely many points with infinitely many nonintersecting line segments. Not all of them can contain critical values. So there exists a good point z_0.

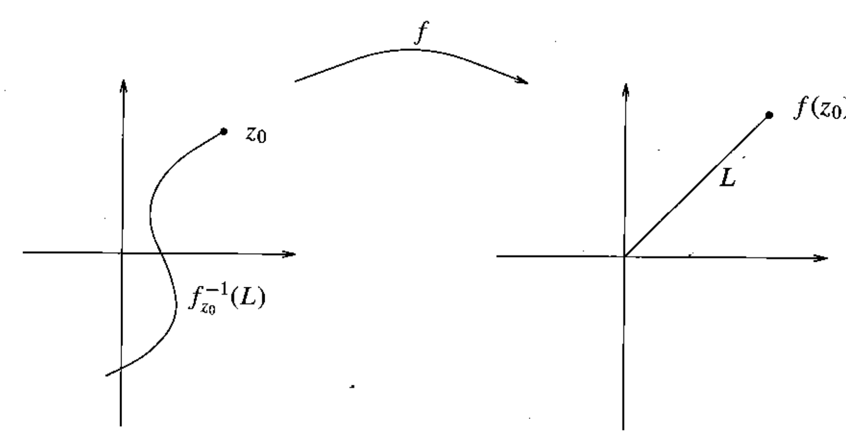

Figure A The line L for z_0 and its pre-image by f_{z_0}.

Let L be as in (9.2) where z_0 is a good point. By the preceding corollary, $f^{-1}(L)$ is closed and bounded.

From the lemmas and the inverse function theorem, take $\delta_z > 0$ so that the inverse $f_z^{-1} : D_{\delta_z}(f(z)) \to \mathbb{C}$ is defined for each $z \in f^{-1}(L)$. Since $f^{-1}(L)$ is closed and bounded there is a $\delta > 0$ independent of z such that δ_z may be choosen bigger than δ. Let n be a positive integer with $n > |f(z_0)|/\delta$.

Define w_i, $i = 0, \ldots, n$, by

$$w_i = \frac{(n-i)f(z_0)}{n}$$

and observe that $|w_i - w_{i-1}| < \delta$, for $i = 1, \ldots, n$; thus

$$w_i \in D_\delta(w_{i-1}).$$

Now one can define inductively

$$z_i = f_{z_{i-1}}^{-1}(w_i), \; i = 1, 2, \ldots, n.$$

Since $f(z_i) = w_i$, $f(z_n) = w_n = 0$. □

Remark 1 Note that

$$f_{z_0}^{-1} : D_\delta(f(z_0)) \to \mathbb{C}$$

may be extended continuously to

$$D_\delta(f(z_0)) \cup D_\delta(f(z_1)) \to \mathbb{C}$$

by uniting it with $f_{z_1}^{-1}$. It can be shown that they agree on the overlap. In this way eventually $f_{z_0}^{-1}$ becomes defined continuously on the δ-neighborhood of L.

9.2 A Homotopy Method

We have shown the existence of zeros of a complex polynomial. We now begin our construction and analysis of an algorithm to locate them. The search problem is as follows

Input: $(\varepsilon, a_0, \ldots, a_d)$, $\varepsilon > 0$, and $f(z) = \sum_{i=0}^{d} a_i z^i$
Output: $x \in \mathbb{C}$ with the property $\left|\sum_{i=0}^{d} a_i x^i\right| < \varepsilon$.

Here $|f(x)| < \varepsilon$ is one possible definition of a solution x. There are many, and the literature on algorithms for this problem is rich. We describe an algorithm and associated constructions which we believe are insightful with widespread implications. This algorithm on one hand is close to and inspired by the proof of the fundamental theorem of algebra in the preceding section. On the other hand, it is a prototype of algorithms called "homotopy methods," "global Newton," and "continuation." The basic increment is one step of Newton's method for an appropriately related polynomial and initial point. We produce a complexity result in two stages, Theorem 2 in this section and Theorem 4 in Section 9.4. They exhibit well the situation described in Section 1.6.

We must be content with determining an approximation. Since Abel and Galois, it has been known that if the degree of f is greater than four, its zeros are not

in general expressible as rational functions of its coefficients, and this remains true even allowing the additional power of extracting roots to our computational abilities.

We are now more precise. Given an accuracy $\varepsilon > 0$, and say $\varepsilon < 1$, $z_0 \in \mathbb{C}$, and f a nonconstant complex polynomial, let $f_t(z) = f(z) - tf(z_0)$, $0 \le t \le 1$. Compare this with Section 9.1. But in what follows we do not exclude the case that 0 is a critical value. Thus let ζ_t be a curve in $\mathbb{C}$, with $f_t(\zeta_t) = 0$, $\zeta_1 = z_0$, and $f_t'(\zeta_t) \neq 0$ for all $t \in (0, 1]$.

We describe a sequence t_i with $t_0 = 1$ and $t_0 > t_1 > \ldots > t_k$ in order to "follow" ζ_t. Let

$$x_0 = z_0\ (= \zeta_1) \quad \text{and} \quad x_i = N_{f_i}(x_{i-1}) \quad \text{for } i \le k, \tag{9.3}$$

where $f_i = f_{t_i}$ (abuse of notation) and $N_{f_i}(x) = x - ((f_i(x))/(f_i'(x))$ is Newton's endomorphism.

Thus one might hope that x_i is defined for all i and that x_i is a good approximation of the zero $\zeta_i = \zeta_{t_i}$ of f_i (another abuse of notation) and even that $|f(x_k)| < \varepsilon$. For $|t_i - t_{i-1}| \le \Delta$ small enough (implying that k is large) this will be the case. Our goal is to achieve this with a good bound on the "complexity" k, as a function of ε, f, z_0.

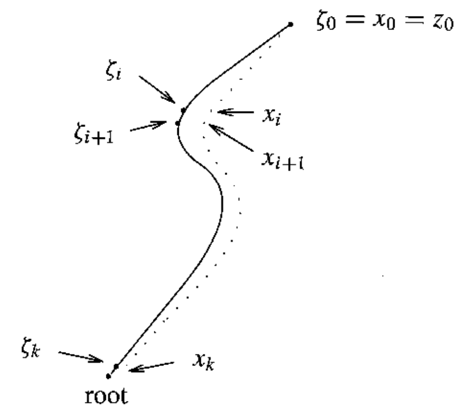

Figure B The curve ζ_t and the dotted path of the x_i that approximate the points $\zeta_i = \zeta_{t_i}$.

The next step is to describe an invariant θ_{f_0,z_0} which gives a measure of how close our path ζ_t comes to a critical point of f. It is some kind of condition number of the pair (z_0, f), as described in Section 1.6.

Definition 2 Given $z_0 \in \mathbb{C}$ and a polynomial f, let the *wedge* $W = W_{f,z_0,\theta}$ of angle θ around $f(z_0)$ be defined by

$$W = \{w \in \mathbb{C} \mid 0 < |w| < 2|f(z_0)|, \arg \frac{w}{f(z_0)} < \theta\}.$$

Here $\arg(w/f(z_0))$ is the angle between w and $f(z_0)$ considered as vectors in $\mathbb{C}$.

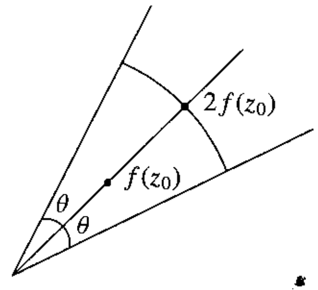

Figure C The wedge of angle θ around $f(z_0)$.

Now we suppose that $f'(z_0) \neq 0$ so that the branch $f_{z_0}^{-1}$ of the inverse of f is well-defined on some disk containing $f(z_0)$. As we saw in the remark after the proof of the fundamental theorem of algebra, $f_{z_0}^{-1}$ may be extended to a larger domain. Let $\theta = \theta_{f,z_0}$ be the largest angle θ with the property that $f_{z_0}^{-1}$ extends to the wedge $W_{f,z_0,\theta} = W$. If, for example, there are no critical values in W, then such an extension always exists. Thus we can speak of the map $f_{z_0}^{-1} : W \to \mathbb{C}$; let $\mathcal{U} = \mathcal{U}_{f,z_0} = f_{z_0}^{-1}(W)$. We may also write $W = W_{f,z_0}$.

Note that there must be a critical value on the boundary of W. However there may be extraneous critical values inside the wedge, which are images of critical points not in $\mathcal{U}$, and perhaps far from ζ_t.

With f, z_0, ε given (as inputs say), let

$$M = M_{f,z_0} = \frac{25}{25 + \sin\theta_{f,z_0}}$$
$$t_i = M^i, \; i = 0, 1, 2, \ldots. \tag{9.4}$$

Theorem 2 *With $\{t_i\}$ as in (9.4), x_i is well-defined for all i in (9.3) and satisfies*

$$|f(x_i)| < 2M^i|f(z_0)| \qquad \textit{for all } i. \tag{9.5}$$

Moreover if

$$k > \left(1 + \frac{25}{\sin\theta_{f,z_0}}\right)\left(\ln|f(z_0)| + \ln\frac{1}{\varepsilon} + 1\right), \tag{9.6}$$

then $|f(x_k)| < \varepsilon$.

Remark 2 Inequality (9.6) is an upper bound for the complexity k. Note that the bound is independent of the degree d of f. Compare this with Section 1.6. This theorem is used subsequently to obtain bounds for k depending on d and the magnitude of the coefficients of f, but not on θ_{f,z_0}.

Proof. We first show how the last statement of the theorem is a consequence of (9.5). From (9.5) our condition to verify is:

$$\frac{1}{M^k} \geq \frac{2|f(z_0)|}{\varepsilon}.$$

Take logarithms to get

$$k \ln\frac{1}{M} > \ln|f(z_0)| + \ln\frac{1}{\varepsilon} + 1.$$

For $0 < s$,

$$\frac{1}{\ln(1+s)} \leq 1 + \frac{1}{s}$$

and the rest follows.

We prove (9.5) by induction on a slightly stronger statement. Let $f_i = f_{t_i}$ (abuse of notation) and consider the statements

(A_i) $\qquad x_i$ is defined and $x_i \in \mathcal{U}_{f,z_0}$; $|f_i(x_i)| \leq (t_i - t_{i+1})|f(z_0)|$

for $i \in \mathbb{N}$. Here the (x_i) are as in (9.3). Note that (A_i) implies (9.5) since $|f_i(x_i)| = |f(x_i) - t_i f(z_0)| \leq (t_i - t_{i+1})|f(z_0)|$ and thus $|f(x_i)| \leq 2t_i|f(z_0)|$. (Recall $t_i = M^i$.) Moreover (A_0) is true since $f_0(x_0) = 0$. The following claim then proves Theorem 2.

Claim *The sentences (A_i) are true for all $i \in \mathbb{N}$.*

As we have already noted (A_0) is true. We prove the claim by induction on i. We proceed in a series of lemmas assuming that (A_i) is true.

To ensure that $x_{i+1} \in \mathcal{U}$, we introduce three curves w_s, v_s, and u_s, for $s \in [0, 1]$ in the following way.

Let w_s be the arc in W_{f,z_0} consisting of two line segments joining:

$$f(x_i) \text{ to } f(\zeta_i) \quad 0 \leq s \leq \frac{1}{2},$$

$$f(\zeta_i) \text{ to } f(\zeta_{i+1}) \quad \frac{1}{2} \leq s \leq 1.$$

Next let $v_s = f_{z_0}^{-1}(w_s)$ lie in $\mathcal{U}$ "over w_s." Note that $v_0 = x_i$, $v_{1/2} = \zeta_i$, and $v_1 = \zeta_{i+1}$. Finally let $u_s = N_{f_{i+1}}(v_s)$, so that u_s is obtained by Newton's method for f_{i+1} applied to v_s for each $s \in [0, 1]$. Thus $u_0 = x_{i+1}$ and $u_1 = \zeta_{i+1}$ (the last since $f_{i+1}(\zeta_{i+1}) = 0$).

For a polynomial f and a complex number z, we defined in the previous chapter a number $\alpha = \alpha(f, z)$ about which we proved in Proposition 2 there that it satisfies the following property: if $\alpha < 1$, then $z' = N_f(z)$ is defined and

$$|f(z')| \leq \frac{\alpha}{1-\alpha}|f(z)|. \tag{9.7}$$

The next proposition comes from Schlicht function theory, the theory of complex analytic functions which are one-to-one on the unit disk. Its proof would carry us too far afield, so we do not give it here. See the last section of this chapter for references for it.

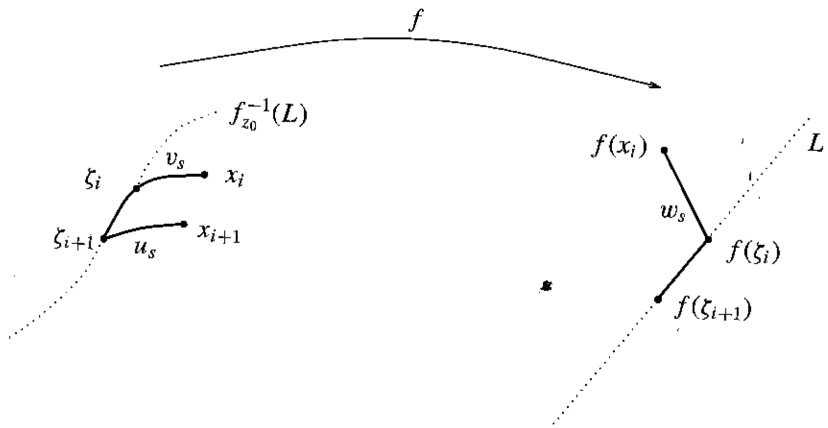

Figure D The curves w_s, v_s, and u_s.

Proposition 1 *For every polynomial f and every $z \in \mathbb{C}$ we have*

$$\alpha(f, z) \leq \frac{4|f(z)|}{r(f_z^{-1})}.$$

□

Here recall that f_z^{-1} is the branch of the inverse of the polynomial f that takes $f(z)$ into z and is defined on a disk of radius r by the implicit function theorem. The largest such r is called the radius of convergence of f_z^{-1} and denoted by $r(f_z^{-1})$.

The following lemmas use the preceding definitions, $\theta = \theta_{f,z_0}$, f_{i+1}, v_s, and M, and suppose that $s \in [0, 1]$.

Lemma 3 (Main Lemma) *If (A_i) holds, then*

$$\alpha(f_{i+1}, v_s) \leq \frac{8(1-M)}{\sin\theta - (1-M)}.$$

Toward the proof of the main lemma, we need some smaller lemmas.

Lemma 4 *If (A_i) holds, then*

$$|f(v_s) - f(\zeta_i)| \leq (t_i - t_{i+1})|f(z_0)|.$$

Proof. This is the same as the inequality $|w_s - f(\zeta_i)| \leq (t_i - t_{i+1})|f(z_0)|$. It is sufficient to check this at the endpoints of the two segments defining w_s. This is quite immediate, using the hypothesis (A_i). □

Recall that $f_i(v_s) = f(v_s) - f(\zeta_i) = f(v_s) - t_i f(z_0)$.

Lemma 5 *If (A_i) holds, then*

$$|f_{i+1}(v_s)| \leq 2(t_i - t_{i+1})|f(z_0)|.$$

Proof.

$$\begin{aligned} f_{i+1}(v_s) &= f(v_s) - t_{i+1} f(z_0) \\ &= f(v_s) - t_i f(z_0) + (t_i - t_{i+1}) f(z_0). \end{aligned}$$

So

$$|f_{i+1}(v_s)| \leq |f(v_s) - t_i f(z_0)| + (t_i - t_{i+1})|f(z_0)|.$$

Now use Lemma 4. □

Lemma 6 *As usual $\theta = \theta_{f,z_0}$. Then if (A_i) holds, one has*

$$r(f_{\zeta_i}^{-1}) \geq |f(\zeta_i)| \sin\theta.$$

Proof. The lemma follows from the diagram (Figure E) and the definitions.

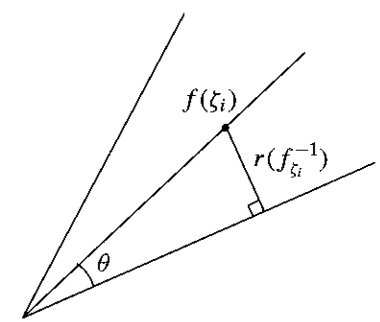

Figure E

□

Lemma 7 *If (A_i) holds, then*

$$r(f_{i+1,v_s}^{-1}) \geq |f(z_0)| M^i (\sin\theta - (1-M)).$$

Proof. From the definitions,

$$r(f_{i+1,v_s}^{-1}) \geq r(f_{\zeta_i}^{-1}) - |f(v_s) - f(\zeta_i)|.$$

Now use Lemmas 4 and 6 noting again that $f(\zeta_i) = t_i f(z_0)$ and $t_i = M^i$. □

Now Lemmas 5 and 7 with Proposition 1 yield Lemma 3. Note that $r(f_{t,z}^{-1})$ is independent of t.

Lemma 8 *Suppose that* (A_i) *is true. If* $\alpha = \alpha(f_{i+1}, v_s)$, $0 \le s \le 1$, *then*

$$\frac{2\alpha}{1-\alpha} \le M.$$

Proof. This is a gentle exercise using Lemma 3 and the definition of M in (9.4). We leave it to the reader. □

Now we are ready to prove the Claim and with it, to finish the proof of Theorem 2.

Proof of the Claim. Let us suppose that (A_i) holds. By inequality 9.7, for $s \in [0,1]$, $|f_{i+1}(u_s)| \le \alpha/(1-\alpha)|f_{i+1}(v_s)|$, α as in Lemma 8. Now by Lemmas 5 and 8 we obtain

$$\begin{aligned}|f_{i+1}(u_s)| &\le M(M^i - M^{i+1})|f(z_0)| \\ &\le (t_{i+1} - t_{i+2})|f(z_0)|.\end{aligned}$$

This estimate yields that $f(u_s) \in W_{f,z_0}$ for all $s \in [0,1]$ and since $u_1 = \zeta_{i+1} \in \mathcal{U} = \mathcal{U}_{f,z_0}$, this means that $u_s \in \mathcal{U}$ for all s. In particular $u_0 = x_{i+1} \in \mathcal{U}$. Moreover for $s = 0$, the last estimate is that of A_{i+1} and with it we finish our proof. □

9.3 Where to Begin the Homotopy

The goal of this section is to make an estimate, which yields many points z with fairly large $\theta_{f,z}$. This estimate is used in the next section to eliminate $\theta_{f,z}$ from the complexity estimate of Theorem 2. Throughout this section, suppose $f(z) = \sum_{i=0}^{d} a_i z^i$, $a_d = 1$, and $|a_i| \le 1$.

Let S_R^1 be the circle of radius R about 0 in $\mathbb{C}$.

Theorem 3 *The set of points* $z \in S_R^1$ *such that* $\theta_{f,z} < a$ *is contained in the union of* $2(d-1)$ *arcs of angle*

$$\frac{2}{d}\left(a + 2\arcsin\frac{1}{R-1}\right), \text{ for } R > 2.$$

To prove Theorem 3 we consider the family of curves in $\mathbb{C}$ defined by taking inverse images under the polynomial f of the family of rays in $\mathbb{C}$. If $w \in \mathbb{C}$ and $w \ne 0$, then the ray through w is $L = L_w = \{\lambda w \mid \lambda \in \mathbb{R},\ \lambda > 0\}$. If $f(z) \ne 0$ and $f'(z) \ne 0$, then by the inverse function theorem f_z^{-1} is defined in a disk B around $f(z)$, where as usual f_z^{-1} takes $f(z)$ to z and $f(f_z^{-1}(z')) = z'$ for $z' \in B$; so f_z^{-1} maps $L_{f(z)} \cap B$ onto a smooth curve c_z through z. The vector $-f(z)$ is tangent to $L_{f(z)}$ so $(f_z^{-1})'(-f(z))$ is tangent to c_z at z. By the inverse function theorem $(f_z^{-1})' = (f'(z))^{-1}$ so the Newton vector $-f'(z)^{-1}f(z)$ is tangent to c_z at z. The c_z fit together into maximal connected curves $\widehat{c_z}$ that partition $\mathbb{C}$ minus the zeros and critical points of f, and $f(\widehat{c_z})$ is contained in the ray $L_{f(z)}$.

If θ is a critical point of f and $f^{(k)}$ is the first nonvanishing derivative of f at θ, then

$$f(z) = f(\theta) + (z-\theta)^k g(z), \tag{9.8}$$

where degree $g = d - k$ and $g(\theta) \ne 0$.

Recall that we may write a nonzero complex number z as $re^{i\phi}$, where r is the modulus $|z|$ of z, and ϕ which is defined up to 2π is the argument of z and is written $\arg(z)$. The representation of a complex number as $re^{i\phi}$ can be accomplished via the exponential mapping

$$e^z = 1 + z + \frac{z^2}{2!} + \ldots + \frac{z^n}{n!} + \ldots,$$

which has the property that

$$e^{z_1+z_2} = e^{z_1}e^{z_2}.$$

In particular if $z = a + bi$ with $a, b \in \mathbb{R}$, then $r = e^a$ and $\phi = b$. Also, $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$. If we denote by ln an inverse function to e^z, then $\arg(z) = \operatorname{Im}\ln(z)$, where $\operatorname{Im}(z)$ is the imaginary part of the complex number z. It then follows from (9.8) that

$$\arg(f(z) - f(\theta)) = k\arg(z-\theta) + \arg(g(z))$$

and that if we restrict f to a small circle $S^1(r,\theta) = \{\theta + re^{i\phi} \mid \phi \in \mathbb{R}\}$ then, taking derivatives, that

$$\frac{d(\arg f(\theta + re^{i\phi}) - f(\theta))}{d\phi} = k + \operatorname{Im}\frac{g'(\theta + re^{i\phi})}{g(\theta + re^{i\phi})} \cdot ire^{i\phi}.$$

We summarize some implications of the previous discussion in a lemma.

Lemma 9 *For* f, θ *as in the preceding with* $f'(\theta) = 0$, $f(\theta) \ne 0$, *and for r small enough,* $\arg(f(\theta + re^{i\phi}) - f(\theta))$ *is monotone in* ϕ *and* f *maps* $S^1(r,\theta)$ k *times around* $f(\theta)$ *intersecting both* $L^+(f(\theta)) = \{\lambda f(\theta) \mid \lambda > 1\}$ *and* $L^-(f(\theta)) = \{\lambda f(\theta) \mid 0 < \lambda < 1\}$ k *times in alternating fashion.*

Thus there are k curves $\widehat{c_z}$ *that map to* $L^+(f(\theta))$ *and k that map to* $L^-(f(\theta))$. *If we orient the curves by* $-f'(z)^{-1}f(z)$, *then those mapping to* $L^+(f(\theta))$ *point into* θ *and those mapping to* $L^-(f(\theta))$ *point out of* θ *(see Figure F).* □

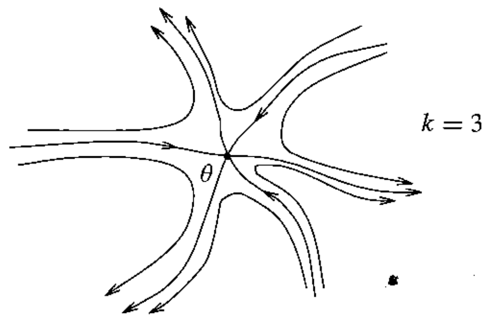

Figure F The curves $\widehat{c_z}$ close to a critical point θ that is not a root.

It remains to consider the family of curves $\widehat{c_z}$ near roots ζ of f. First we recall that if $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$ where x_i, y_i are real for $i = 1, 2$, then

$$\mathrm{Re}(\bar{z}_1 z_2) = \mathrm{Re}(z_1 \bar{z}_2) = x_1x_2 + y_1y_2$$

and the last is the usual dot product of z_1 and z_2 considered as vectors in $\mathbb{R}^2$ which we denote in the rest of this chapter by $\langle z_1, z_2\rangle$. The next lemma helps to understand the overall geometric picture.

Lemma 10 *One has*

$$-\mathrm{grad}\left(\frac{1}{2}|f(z)|^2\right) = -f(z)\overline{f'(z)}.$$

Proof. For any $v \in \mathbb{C}$

$$\begin{aligned}\langle -\mathrm{grad}\left(\frac{1}{2}|f(z)|^2\right), v\rangle &= -\frac{1}{2}(\langle f(z), f(z)'v\rangle + \langle f'(z)v, f(z)\rangle)\\ &= -\langle f(z), f'(z)v\rangle = -f(z)\overline{f'(z)}\bar{v}\\ &= \langle -f(z)\overline{f'(z)}, v\rangle.\end{aligned}$$

□

From Lemma 10 it follows that the Newton vector $-f'(z)^{-1}f(z)$ and $-\mathrm{grad}\left(\frac{1}{2}|f(z)|^2\right)$ differ by a positive real multiple since $|f'(z)|^2 = f'(z)\overline{f'(z)}$. Thus the curves $\widehat{c_z}$ are tangent to $-\mathrm{grad}\left(\frac{1}{2}|f(z)|^2\right)$ and orthogonal to the level curves of $|f(z)|^2$. Near a zero ζ these curves limit on ζ and their orientation points toward ζ (toward Figure G).

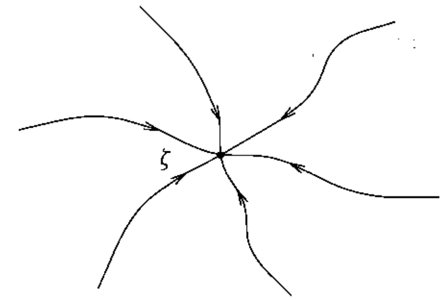

Figure G The curves $\widehat{c_z}$ close to a multiple root ζ.

Now we show that the curves $\widehat{c_z}$ cross and enter into any large circle centered at 0. This leads to the proof of Theorem 3.

Lemma 11 *Let $f(z) = z^d + a_{d-1}z^{d-1} + \ldots + a_0$ be a complex polynomial with $|a_i| \le 1$ for $0 \le i \le d-1$. Let $R \ge 2$. Then for any $z \in S^1_R$,*

$$\mathrm{Re}\left(\bar{z}\frac{f(z)}{f'(z)}\right) > 0.$$

Proof. Let $w_1, \ldots, w_d$ be the roots of f; then $|w_i| < 2$ by Lemma 2. The w_i are in the half-plane defined by the tangent line to the circle at z; that is, $\mathrm{Re}(\bar{z}w_i) < \mathrm{Re}(\bar{z}z)$.

So $\mathrm{Re}(\bar{z}(z-w_i)) > 0$ for all i, $1 \le i \le d$ and $\mathrm{Re}(z/(z-w_i)) > 0$ for all i, $1 \le i \le d$ and $\mathrm{Re}\left(z\sum_{i=1}^d 1/(z-w_i)\right) > 0$; so

$$\mathrm{Re}\left(z\frac{f'(z)}{f(z)}\right) > 0 \text{ and } \mathrm{Re}\left(\bar{z}\frac{f(z)}{f'(z)}\right) > 0.$$

(In the proof we have used

$$\mathrm{Re}(\bar{u}v) > 0 \text{ iff } \mathrm{Re}(u\bar{v}) > 0 \text{ iff } |v|^2\,\mathrm{Re}\left(u\frac{1}{v}\right) > 0.)$$

□

From the last lemma if $z \in S^1_R$, then the inner product satisfies

$$\langle z, \frac{-f(z)}{f'(z)}\rangle = \mathrm{Re}\left(\frac{-\bar{z}f(z)}{f'(z)}\right) < 0.$$

Lemma 12 *Suppose $f(z) = z^d + a_{d-1}z^{d-1} + \ldots + a_0$ with $|a_i| \le 1$ for $0 \le i \le d-1$. Let $z_1, z_2 \in S^1_R$ such that*

$$\beta \le \left|\arg\frac{z_1}{z_2}\right|.$$

Then

$$\left|\arg \frac{f(z_1)}{f(z_2)}\right| \geq d\beta - 2\arcsin\frac{1}{R-1}.$$

Proof.

$$\frac{f(z)}{z^d} = 1 + \frac{a_{d-1}z^{d-1} + \ldots + a_0}{z^d}$$

and for $z \in S_R^1$

$$\frac{\left|\sum_{j=0}^{d-1} a_j z^j\right|}{|z^d|} \leq \frac{\sum_{j=0}^{d-1} R^j}{R^d} = \frac{R^d - 1}{(R-1)R^d} < \frac{1}{R-1}.$$

Thus $f(z)/z^d$ is inside the circle of radius $1/(R-1)$ centered by 1 (see Figure H), and

$$\left|\arg \frac{f(z)}{z^d}\right| \leq \arcsin\frac{1}{R-1}.$$

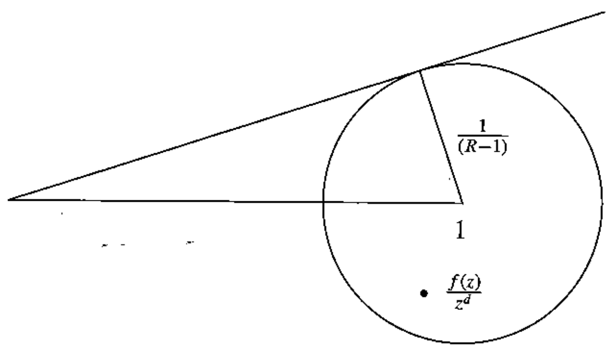

Figure H The location of $f(z)/z^d$.

Now

$$\frac{f(z_1)}{f(z_2)} = \frac{z_1^d(f(z_1)/z_1^d)}{z_2^d(f(z_2)/z_2^d)} = \left(\frac{z_1}{z_2}\right)^d \frac{f(z_1)/z_1^d}{f(z_2)/z_2^d}$$

and the lemma follows from the fact that the argument of a product is additive and the triangle inequality. □

Let θ be a critical point of f. Let L be the ray through $f(\theta)$, $L = \{\lambda f(\theta) \mid \lambda > 0\}$, and Σ_θ the component of $f^{-1}(L)$ that contains θ. Let

$$\Sigma = \bigcup_{\theta \mid f'(\theta)=0} \Sigma_\theta.$$

Lemma 13 *If $R \geq 2$ and f is a monic polynomial of degree d whose coefficients are bounded by 1 in absolute value, then $\Sigma \cap S_R^1$ is a set of at most $2(d-1)$ points.*

Proof. Recall that the curves $\widehat{c}_z$ which are inverse images of rays by f are tangent to the point $(-f(z))/f'(z)$ which by Lemma 11 is transversal to S_R^1 for any $R \geq 2$. It follows that for any fixed θ_i such that $f'(\theta_i) = 0$, $\sum_{\theta_i} \cap S_R^1$ consists of at most $k_i + 1$ points where k_i is the multiplicity of θ_i *as a root of* f'. Thus

$$\sum_{\theta_i \mid f'(\theta_i)=0} k_i = d - 1 \qquad \text{and} \qquad \sum_{\theta_i \mid f'(\theta_i)=0} (k_i + 1) \leq 2(d-1).$$

Equality is only obtained here if each critical point θ_i is a simple zero of f', that is, if each $k_i = 1$. □

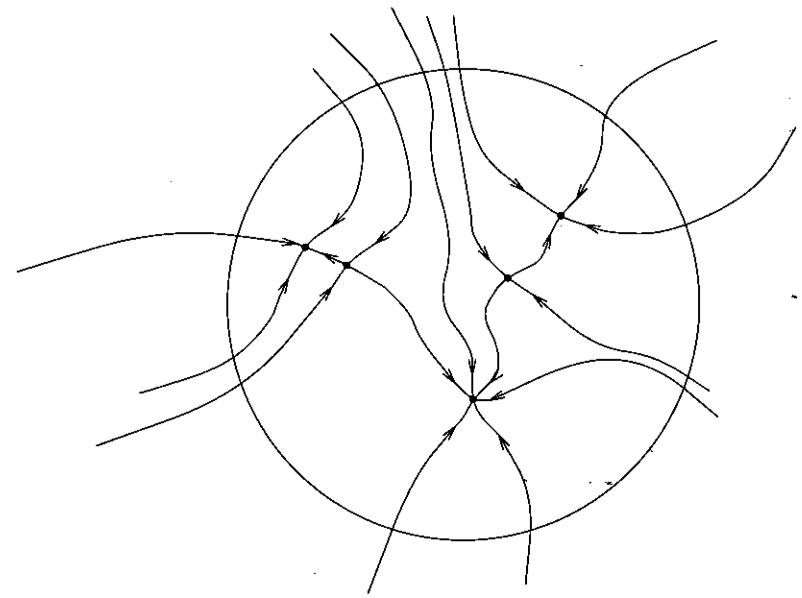

Figure I An overall geometric picture.

Proof of Theorem 3. Let $x_1, \ldots, x_l$ with $0 \leq l \leq 2(d-1)$ be the set of points, $\Sigma \cap S_R^1$, in Lemma 13. If for all i, $1 \leq i \leq l$,

$$\left|\arg \frac{z}{x_i}\right| \geq \frac{1}{d}\left(a + 2 \operatorname{arc\,sin} \frac{1}{R-1}\right),$$

then by Lemma 12

$$\left|\arg \frac{f(z)}{f(x_i)}\right| \geq a$$

and $\theta_{f,z} \geq a$. So if $\theta_{f,z} < a$, then there exist some x_i, $i = 1, \ldots, l$, such that

$$\left|\arg \frac{z}{x_i}\right| < \frac{1}{d}\left(a + 2 \operatorname{arc\,sin} \frac{1}{R-1}\right)$$

and this proves Theorem 3. □

9.4 The Algorithm and Its Complexity

The method of Theorem 2 for finding roots depends on a choice of z_0. Using Theorem 3, we show how to make that choice intelligently. This leads to an algorithm with a good complexity bound for approximating a zero of a polynomial.

Now motivated by Theorem 3, let a and $R > 2$ satisfy

$$a + 2\arcsin\frac{1}{R-1} < \frac{\pi}{4}. \tag{9.9}$$

Then using Theorem 3, we may conclude that the set of points $z \in S_R^1$ with $\theta_{f,z} < a$ is contained in the union of $2(d-1)$ arcs, each of angle less than $\pi/2d$. Thus no two of the $4d$ evenly spaced points on S_R^1, $Re^{(2\pi j/4d)\sqrt{-1}}$, $j = 0, \ldots, 4d-1$ lie in the same one of these $2(d-1)$ arcs. So at least one-half of these points have the associated $\theta_{f,z} \geq a$.

Let us now consider M as given in (9.4) and k the smallest integer satisfying

$$k \geq \left(1 + \frac{25}{\sin a}\right)\left(d\ln R + \ln\frac{1}{\varepsilon} + 1\right). \tag{9.10}$$

For these a, R, M, and k consider the algorithm:

Input (ε, f), $f(z) = z^d + a_{d-1}z^d + \ldots + a_0$, $|a_i| \leq 1$

(1) Let k be as in (9.10).

(2) Choose $j \in \{0, 1, \ldots, 4d-1\}$ not previously chosen.

(3) Let $x_0 = Re^{(2\pi j\sqrt{-1})/4d}$. Let $x_i = N_{f_i}(x_{i-1})$, for $i = 1, \ldots, k$, be as in Theorem 2 (using $t_i = M^i$, etc.).

(4) If $|f(x_k)| < \varepsilon$, output x_k, else go to (2).

Note that Theorems 2 and 3 ensure that the algorithm is correct in the sense that it halts in Step (4). Thus, it performs at most $4d$ iterations of its main loop. On the other hand, Step (3) performs exactly k evaluations of the Newton's endomorphism. We can summarize this in our main theorem.

Theorem 4 *The algorithm just described returns an approximate root of f performing at most $4kd$ evaluations of the Newton's endomorphism k as in (9.10). If j in Step (2) is chosen at random, then on the average it performs only $2k$ such evaluations.* □

Remark 3 One may choose $a = 0.628 < \pi/5$, $R = 12.745$ (that satisfies $2\arcsin 1/(R-1) < \pi/20$). This choice yields for k the smallest integer greater than $43.56(2.55d - \ln\varepsilon + 1)$.

Remark 4 One may obtain the requisite number of arithmetic operations (and comparisons) by multiplying the number of evaluations of the Newton's endomorphism by the number of operations done at each one of these evaluations. This latter number is less than $3d$, and hence the complexity estimates of Theorem 4 get multiplied by $3d$. The total number of arithmetic operations is therefore —for the a and R of the preceding remark— bounded by

$$514d^2(2.55d + \ln\frac{1}{\varepsilon} + 1)$$

and thus within $O(d^2(d + \ln 1/\varepsilon))$.

Remark 5 The numbers $e^{(2\pi j\sqrt{-1})/4d}$ in Step (3) are not represented by rational numbers. But that is dealt with by simple approximations, and a little thought. The complexity estimates are not significantly affected.

Remark 6 The algorithm used in the first half of Theorem 4 is one in the sense of our machines over $\mathbb{R}$ defined in Part I, taking into account Remark 5. Then our result is a prototype of a polynomial time bound of a numerical analysis problem as suggested in Section 1.6.

Remark 7 The algorithm in the second part of Theorem 4 uses a random choice. In terms of machines over $\mathbb{R}$, this could be encompassed by adding an appropriate type of node, one which yields a random choice. Note the speedup factor of d.

Remark 8 Suppose one is given as input ε and $f(z) = \sum_0^d a_i z^i$, $a_d \neq 0$ without the normalizing conditions on the coefficients. Then one uses a preliminary procedure explicit in the proof of Lemma 2 to obtain a normalized polynomial g. In order to obtain an element x satisfying $|f(x)| < \varepsilon$ it is enough to obtain a y satisfying that

$$g(y) < \frac{\varepsilon}{a_d b^d}, \qquad \text{where} \qquad b = \max_{1\leq i\leq d}\left(\left|\frac{a_{d-i}}{a_d}\right|^{1/i}, 1\right).$$

Therefore, the $\ln(1/\varepsilon)$ of the complexity bound in Remark 4 now becomes $\ln(1/\varepsilon) + d\ln b + |\ln|a_d||$.

Remark 9 Note that in the implementation of this algorithm it is not necessary to store all the values x_i in Step (3). As a matter of fact, one can initialize a variable x with the value of x_0 and update this value for $i = 1$ to k.

Remark 10 The same algorithm slightly modified yields all the roots of a polynomial with a corresponding complexity analysis.

9.5 Additional Comments and Bibliographical Remarks

Proofs of the fundamental theorem of algebra are legion. The one here follows [Smale 1986b] closely but has roots in the thesis of Gauss (see [Smale 1981] for historical remarks). The statement in Remark 1 follows immediately from considerations proved in elementary complex variable books (see, e.g., Lang [1993b]).

The complexity result of Section 9.4 is a distillation of the efforts of Smale [1981], Shub and Smale [1985, 1986], and Smale [1986b]. However, the proof in Section 9.2 via Proposition 2 of Chapter 8 is an improvement over the previous accounts. Proposition 1 is derived very quickly in [Smale 1981] from a more difficult theorem of Loewner (and Loewner's theorem has four proofs in [Schober 1980]).

For Remark 10 on finding all the roots, one can see [Shub and Smale 1985; Shub and Smale 1986].

There is some history of complexity results for algorithms implementing the fundamental theorem of algebra as in [Dejon and Henrici 1969; Ostrowski 1973; Henrici 1977; Schönhage 1982]. Sharper results than Theorem 4 with different (and more complicated) algorithms are, for example, in [Renegar 1987b; Neff 1994; Neff and Reif 1996; Pan 1995, TA].

The main algorithm of this chapter is a homotopy or continuation method. This is the method of choice in recent times for solving systems of equations by numerical techniques. Here is some of the literature: [Kellog, Li, and Yorke 1976; Eaves and Scarf 1976; Smale 1976; Keller 1978; Garcia and Gould 1980; Hirsch and Smale 1979; Garcia and Zangwill 1981; Allgower and Georg 1990, 1993, 1997; Morgan 1987; Li, Sauer, and Yorke 1987]. These works mostly do not deal with complexity considerations. Later ones which do are [Renegar 1987a] and [Shub and Smale 1993a, 1993b, 1993c, 1996, 1994].

A main alternative class of methods for solving polynomial systems of equations are based on algebra (elimination theory, Gröbner bases, effective Nullstellensatz). We do not try to deal with them here.

10

Bézout's Theorem

Bézout's Theorem is the n-dimensional generalization of the Fundamental Theorem of Algebra. It "counts" the number of solutions of a system of n complex polynomial equations in n-unknowns. It is the goal of this chapter to prove Bézout's Theorem. In Chapter 16 we use Bézout's Theorem as a tool to derive geometric upper bounds on the number of connected components of semi-algebraic sets and complexity-theoretic lower bounds on some problems such as the Knapsack.

We proceed rather leisurely. First we revisit the Fundamental Theorem of Algebra and prove it again in such a way that the proof goes over to prove Bézout's Theorem in an extended geometric form. The proof proceeds by constructing a homotopy from a system with known roots. Most of the effort goes into showing that, generally speaking, pairs of systems of equations may be joined by a path that avoids the set of systems of equations with multiple roots, the discriminant variety. As in the proof of the Fundamental Theorem of Algebra, the homotopy gives rise to a path of solutions that may be followed numerically by Newton's method. We defer a complexity analysis of this approach to finding roots of systems of equations to Chapter 14.

10.1 The Fundamental Theorem of Algebra Revisited

Let $f(z) = a_d z^d + \ldots + a_0$ with $a_d \neq 0$ a complex polynomial. From the Fundamental Theorem of Algebra we know that there is a root r_1 of $f(z)$ and $a_d^{-1} f(z) = (z - r_1)g(z)$, where the degree of g is $d - 1$. Thus via induction there are d complex numbers $r_1, \ldots, r_d$ such that $f(z) = a_d(z - r_1)(z - r_2) \ldots (z - r_d)$.

There may be fewer than d distinct numbers in the list $r_1, \ldots, r_d$. The *multiplicity* of a root is the number of times it appears in the list $r_1, \ldots, r_d$ of roots. Thus another version of the Fundamental Theorem of Algebra asserts that a complex polynomial f has d roots, with the provisos that the roots are counted with multiplicity and $a_d \neq 0$.

In order to better understand the proviso $a_d \neq 0$ let us consider in some more detail the case $d = 2$, with real coefficients; that is,

$$f(z) = az^2 + bz + c \qquad a \neq 0.$$

It is well known that the two roots of this equation are

$$\zeta_+ = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \qquad \text{and} \qquad \zeta_- = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

If we fix b and c with $b > 0$ and we let a tend to zero we find that ζ_+ tends to $-c/b$ and ζ_- "escapes" to infinity. One way to avoid this type of escape is to embed the space where the roots are found in a compact space where escape is impossible. The space obtained after this compactification, called projective space, contains some new points that are "at infinity." In one variable the addition of a single point is sufficient to capture the behavior of a polynomial at infinity. When dealing with systems of polynomials in several variables, the set of points at infinity in multidimensional projective space will be rich enough to reflect the direction that finite roots take in their escape to infinity.

Now let us be more precise about the way this compactification is done and the form that the Fundamental Theorem of Algebra takes in this compact space.

If $f(z, w) = a_d z^d + a_{d-1} z^{d-1} w + \ldots + a_0 w^d$ and some $a_i \neq 0$, we say that f is an *homogeneous polynomial* of z and w of degree d. We are no longer assuming $a_d \neq 0$. If (z, w) is a zero of an homogeneous polynomial, then so is $(\lambda z, \lambda w)$ for any $\lambda \in \mathbb{C}$. We express this in a lemma.

Lemma 1 *If $f(z, w)$ is an homogeneous polynomial of degree d, then*

$$f(\lambda z, \lambda w) = \lambda^d f(z, w) \text{ for all } \lambda \in \mathbb{C},$$

and hence if $f(z, w) = 0$, then $f(\lambda z, \lambda w) = 0$ for all $\lambda \in \mathbb{C}$.

We say that a (nontrivial) line $\{(\lambda z, \lambda w) \mid x \in \mathbb{C}\}$ is a *solution* of $f(z, w) = 0$ if

$$f(\lambda z, \lambda w) = 0 \text{ for all } \lambda \in \mathbb{C}.$$

The fundamental theorem of algebra now takes the following form.

Theorem 1 (The Fundamental Theorem of Algebra—Version 2.) *Let $f(z, w) = a_d z^d + a_{d-1} z^{d-1} w + \ldots + a_0 w^d$ be a nonzero complex homogeneous polynomial of degree d. Then f has d complex solutions provided the solutions are counted with multiplicity.*

Theorem 1 is intimately related to Theorem 1 of Chapter 9. Given a polynomial $f(z) = a_d z^d + a_{d-1} z^{d-1} + \ldots + a_0$ its *homogenization* is the homogeneous polynomial $\widehat{f}(z, w) = a_d z^d + a_{d-1} z^{d-1} w + \ldots + a_0 w^d$. If $f(z) = 0$, then $\widehat{f}(z, 1) = 0$, and if $\widehat{f}(z, w) = 0$, then $\widehat{f}((z/w), 1) = 0$ and $f(z/w) = 0$ as long as $w \neq 0$.

The zeros of $\widehat{f}(z, w)$ with $w = 0$ but $z \neq 0$ are called *zeros at infinity* for f. They occur only if $a_d = 0$. Thus except for zeros at infinity the zeros of f are given by intersecting the line $w = 1$ with the solution lines of $\widehat{f}$ (see Figure A).

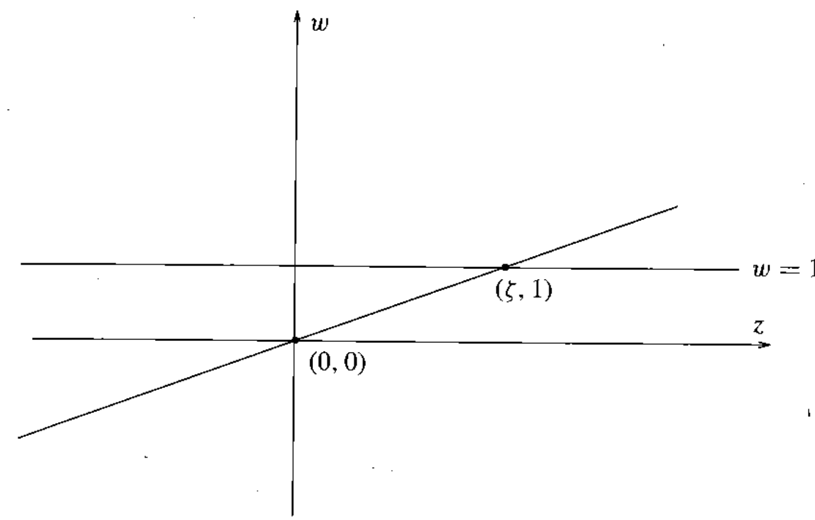

Figure A A zero line of $\widehat{f}$, corresponding to the zero ζ of f.

If we let $\mathcal{P}_d$ be the vector space of all polynomials of degree less than or equal to d, and $\mathcal{H}_d$ the vector space of all homogeneous polynomials of degree d, then the map given by homogenization

$$\begin{array}{cccc} \wedge : & \mathcal{P}_d & \to & \mathcal{H}_d \\ & f & \to & \widehat{f} \end{array}$$

is a linear isomorphism. The inverse is obtained by setting $w = 1$. Both vector spaces $\mathcal{P}_d$ and $\mathcal{H}_d$ have dimension $(d + 1)$.

Using these ideas, it is not difficult to derive the second version of the Fundamental Theorem of Algebra from the first one in the preceding chapter. Notwithstanding, we devote the next three sections to a different proof of Theorem 1. The reason is that the ideas behind this latter proof naturally extend to polynomial systems in several variables and, eventually, they lead to an algorithm for computing approximate zeros of these systems.

Version 2 of the Fundamental Theorem of Algebra asserts that the homogeneous complex polynomial f of degree d has d solutions which are lines through zero in $\mathbb{C}^2$. Thus it is natural to define the space of lines through zero. We do this in general in the next section.

10.2 Projective Space

Definition 1 Let $\mathbb{C}^{n+1}$ be the $(n+1)$-dimensional complex Cartesian space so that $\mathbb{C}^{n+1} = \{(x_0, \ldots, x_n) \mid x_i \in \mathbb{C} \text{ for } i = 0, \ldots, n\}$. Then *n-dimensional complex projective space* $\mathbb{P}(\mathbb{C}^{n+1})$ is the set of one-dimensional vector subspaces of $\mathbb{C}^{n+1}$ (i.e., complex lines through zero) or we may say $\mathbb{P}(\mathbb{C}^{n+1})$ is the set of $x \in \mathbb{C}^{n+1}$, where we identify x with λx whenever λ is a nonzero complex number.

With this definition the map

$$\begin{aligned} \mathbb{C}^n &\to \mathbb{P}(\mathbb{C}^{n+1}) \\ x &\to (1, x) \end{aligned}$$

is an embedding of $\mathbb{C}^n$ into $\mathbb{P}(\mathbb{C}^{n+1})$ whose image is $\mathbb{P}(\mathbb{C}^{n+1})$ less the points of the form $(0, x)$ with $x \in \mathbb{C}^n - \{0\}$ (or the "points at ∞"). The addition of these points at infinity makes $\mathbb{P}(\mathbb{C}^{n+1})$ compact. "Escape to infinity" is no longer a problem. "Infinity" is in our space.

In fact, one is not actually using the coordinate structure so that if W is a finite-dimensional complex vector space, there is a naturally *associated projective space*, $\mathbb{P}(W)$. Thus, $\mathbb{P}(W)$ is defined as the set of nontrivial $w \in W$, where $v, w \in W$ are identified when $v = \lambda w$ for $\lambda \in \mathbb{C} - \{0\}$.

In general, there is a natural map $Q : W - \{0\} \to \mathbb{P}(W)$ which takes a nonzero vector v to the unique one-dimensional vector subspace in which it lies. Thus, $Q^{-1}(x)$ for $x \in \mathbb{P}(W)$ is a line through 0 minus $\{0\}$.

An *Hermitian product* on a complex vector space W is a map

$$\begin{aligned} W \times W &\to \mathbb{C} \\ (v, w) &\quad \langle v, w \rangle \end{aligned}$$

satisfying, for all $u, v, w \in W$, and $\lambda \in \mathbb{C}$,

1. $\langle v, w \rangle = \overline{\langle w, v \rangle}$,
2. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$,
3. $\langle \lambda v, w \rangle = \lambda \langle v, w \rangle$,
4. $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0$ implies $v = 0$.

The standard Hermitian product on $\mathbb{C}^{n+1}$ is given by

$$\langle v, w \rangle = \sum_{i=0}^{n} v_i \bar{w}_i,$$

where $v = (v_0, \ldots, v_n)$ and $w = (w_0, \ldots, w_n)$. An Hermitian product on a vector space W determines many properties of the vector space W which are related to it. We mention a few here which we use.

Two vectors $v, w \in W$ are *orthogonal* if $\langle v, w \rangle = 0$. It follows that if $v \in W$ and $v \neq 0$, then the set of vectors orthogonal to v,

$$T_v = \{w \in W \mid \langle v, w \rangle = 0\},$$

is a vector subspace of W. If W has dimension n, then T_v has dimension $n - 1$. A basis $w_1, \ldots, w_n$ of W is called orthogonal if $\langle w_i, w_j \rangle = 0$ for $i \neq j$. Any finite-dimensional vector space has an orthogonal basis.

An Hermitian product also determines a norm on W by

$$\|w\| = \langle w, w \rangle^{1/2}.$$

Once we have a norm on W we have the unit sphere $S_1 \subset W$,

$$S_1 = \{w \in W \mid \|w\| = 1\}.$$

An orthogonal basis $w_1, \ldots, w_n$ of W is called *orthonormal* if $\|w_i\| = 1$ for all $i = 1, \ldots, n$. If $v_1, \ldots, v_n \in V$ and $w_1, \ldots, w_n \in W$ are the orthonormal basis of V and W, respectively, then the unique linear map L from V to W taking v_i to w_i for $i = 1, \ldots, n$ is an isomorphism and preserves lengths; that is, $\|L(v)\| = \|v\|$ for all $v \in V$. Thus, L maps the unit sphere in V to the unit sphere in W and they are homeomorphic. Since the unit sphere S^{2n-1} in $\mathbb{C}^n$ is compact it follows that the unit sphere in any finite-dimensional vector space is compact.

We denote by ρ the restriction of Q to S_1,

$$\rho : S_1 \to \mathbb{P}(W).$$

Since ρ is continuous it follows that $\mathbb{P}(W)$ is also compact. For $x \in \mathbb{P}(W)$, $\rho^{-1}(x)$ is the unit circle in $Q^{-1}(x)$. Another way to express this is that the unit circle S^1 in $\mathbb{C}$ acts on S_1 and the quotient by this action is $\mathbb{P}(W)$.

Let L be a line through 0 in W and let $0 \neq w \in L$. The set

$$A_w = w + T_w$$

is an affine subspace of W. Since T_w has dimension $n - 1$ and w is orthogonal to T_w it follows that any line L' close to L intersects the space A_w in a unique point. If we let $\phi_w(L')$ be this point, then ϕ_w is defined in a neighborhood U_L of L in $\mathbb{P}(W)$ and is a homeomorphism there. Let $\psi_w = \phi_w^{-1}$. Then ψ_w is defined in a neighborhood V_w of w in A_w,

$$\psi_w : V_w \to \mathbb{P}(W).$$

In the language of differentiable manifolds the ψ_w are *charts* for $\mathbb{P}(W)$.

A map g defined on U_L taking values in a differentiable manifold N, $g : U_L \to N$, is *differentiable at* L if $g \circ \psi_w : V_w \to N$ is differentiable at w. In that case the derivative of $g \circ \psi_w$ is defined on T_w. Now just as $\mathbb{P}(W)$ is defined as equivalence classes of points $x \in W$ where we identify x with λx for $\lambda \in \mathbb{C}, \lambda \neq 0$, the tangent bundle of $\mathbb{P}(W)$ is defined as equivalence classes $(x, v) \in W \times W$ where $v \in T_x$ and (x, v) is equivalent to $(\lambda x, \lambda v)$ for $\lambda \in \mathbb{C}, \lambda \neq 0$. The necessity for the scaling in the vector v is evident in Figure B or from the chain rule.

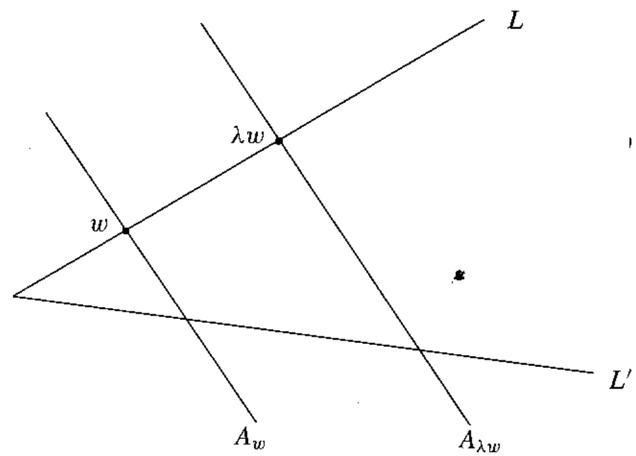

Figure B The vector from L' to L in $A_{\lambda w}$ is λ times the vector from L' to L in A_w.

Remark 1 The definition of complex projective space we have given does not hinge on any special feature of the complex field and can be extended to vector spaces over arbitrary fields. In the case of the field of real numbers, the rest of the discussion holds as well replacing the Hermitian structure in $\mathbb{C}^{n+1}$ by the standard dot product in $\mathbb{R}^{n+1}$. We meet real projective spaces in Chapter 16.

We meet differentiable manifolds at several points in this book. We close this section with a formal definition of differentiable manifolds and maps. For a detailed treatment of such concepts see the references in Section 10.6.

Definition 2 A topological space M is an *n-dimensional differentiable manifold* if there is a set A, open sets $U_a \subset \mathbb{R}^n$ for $a \in A$, and functions $\psi_a : U_a \to M$ such that

1. $\psi_a : U_a \to M$ is a homeomorphism from U_a onto its image for all $a \in A$,
2. $\cup_{a \in A} \psi_a(U_a) = M$, and
3. $\psi_b^{-1}\psi_a$ is a smooth map on the open set in which it is defined for all $a, b \in A$.

The maps ψ_a are called *charts*. The set of charts $\{\psi_a\}$ for $a \in A$ is called an *atlas*. An atlas $\{\psi_a\}_{a \in A}$ is *maximal* if any homeomorphism $\Phi : U \to M$ with $U \subset \mathbb{R}^n$ an open subset such that $\psi_a \Phi$ and $\Phi \psi_a$ are smooth maps on the open sets on which they are defined for all $a \in A$ is already an element of $\{\psi_a\}_{a \in A}$.

Any atlas is contained in a maximal atlas, which defines a *differentiable structure* on M.

If M and N are differentiable manifolds, then $f : M \to N$ is *differentiable* if $\psi_a^{-1} f \psi_b$ is differentiable for all $\{\psi_a\}_{a \in A}$ and $\{\psi_b\}_{b \in B}$ atlases of M and N, respectively.

Let $\mathbb{R}^k \subset \mathbb{R}^n$ be the vector subspace of $\mathbb{R}^n$ consisting of those vectors whose last $(n-k)$ coordinates are 0.

Definition 3 A subset N of the n-dimensional differentiable manifold M is a *submanifold* of M if there are charts for M, $\psi_b : U \to M$ for $b \in B$, such that

1. $\psi_b : (U_b \cap \mathbb{R}^k) \subset N$ for all $b \in B$, and
2. $(\psi_b | U_b \cap \mathbb{R}^k) : U_b \cap \mathbb{R}^k \to N$ for $b \in B$ is an atlas for N.

10.3 The Varieties V, Σ', and Σ

Define $\mathcal{H}_d$ as the set of all homogeneous polynomials in $n+1$ variables of degree d. Let the vector of degrees $(d) = (d_1, \ldots, d_k)$, fix $n \geq 2$, and let $\mathcal{H}_{(d)} = \mathcal{H}_{d_1} \times \ldots \times \mathcal{H}_{d_k}$, so $\mathcal{H}_{(d)}$ is the space of all polynomial mappings $f = (f_1, \ldots, f_k)$, $f : \mathbb{C}^{n+1} \to \mathbb{C}^k$ such that $f_i \in \mathcal{H}_{d_i}$ for all i. We may define the *solution variety*

$$V = \{(f, x) \in \mathbb{P}(\mathcal{H}_{(d)}) \times \mathbb{P}(\mathbb{C}^{n+1}) \mid f(x) = 0\}.$$

Then V is well-defined. It plays a crucial role in our proof. Here we use $f \in \mathbb{P}(\mathcal{H}_{(d)})$ and $x \in \mathbb{P}(\mathbb{C}^{n+1})$ by choosing representatives "f" $\in \mathcal{H}_{(d)}$ and "x" $\in \mathbb{C}^{n+1}$. This is legitimate essentially because our definitions are so natural.

We now establish that V is a smooth and connected submanifold of $\mathbb{P}(\mathcal{H}_{(d)}) \times \mathbb{P}(\mathbb{C}^{n+1})$ of complex codimension k whose tangent space at the point (f, x) is the vector space of (h, w), where $h \in T_f$, $w \in T_x$, and $h(x) + Df(w) = 0$.

Recall that if $f : M \to N$ is a differentiable map, then $q \in N$ is a *regular value* of f if and only if for every $p \in f^{-1}(q)$, the derivative of f at p is a surjective linear map. The derivative maps the tangent space of M at p to the tangent space of N at q. If $M = \mathbb{C}^m$ and $N = \mathbb{C}^n$, the tangent spaces are $\mathbb{C}^m$ and $\mathbb{C}^n$, respectively. For the manifolds and smooth varieties we consider, we usually try to describe the tangent space.

Proposition 1 *The solution variety V is a smooth connected subvariety of $\mathbb{P}(\mathcal{H}_{(d)}) \times \mathbb{P}(\mathbb{C}^{n+1})$ of codimension k. The tangent space to V at (f, x) is the vector space of (h, w) where $h \in T_f$, $w \in T_x$, and $h(x) + Df(x)(w) = 0$.*

Proof. Let the evaluation map ev be defined by

$$\begin{array}{rccc} ev: & \mathcal{H}_{(d)} \times \mathbb{C}^{n+1} & \longrightarrow & \mathbb{C}^k \\ & (f, x) & \to & f(x). \end{array}$$

Then the derivative of the evaluation map at (f, x) applied to (h, w) is $D(ev_{(f,x)})(h, w) = h(x) + Df(x)w$. Taking $w = 0$ and varying h only we see that 0 is a regular value of ev. Let $V' = \{(f, x) \in (\mathcal{H}_{(d)} - \{0\}) \times (\mathbb{C}^{n+1} - \{0\}) \mid f(x) = 0\}$. Then V' is smooth by the inverse function theorem, its codimension is k, and its tangent space at (f, x) is the vector space of (h, w) such that $h(x) + Df(x)w = 0$.

We also claim that V' is connected. For $x \in \mathbb{C}^{n+1} - \{0\}$ let $V'_x = \{(f, x) \in V'\}$. The elements $f \in \mathcal{H}_{(d)} - \{0\}$ appearing in the first coordinate of V'_x form a linear subspace of $\mathcal{H}_{(d)} - \{0\}$ of codimension k. Hence V'_x is connected. Now we show that given $x, x' \neq 0$ there is a path connecting V'_x to $V'_{x'}$. Let $f \in V'_x$ and let L_t be a family of invertible linear maps $L_t : \mathbb{C}^{n+1} \to \mathbb{C}^{n+1}$ for $0 \leq t \leq 1$ such that the association $t \to L_t$ is continuous, $L_0(x) = x$, and $L_1(x) = x'$. Then $(f_t, x_t) = (f \circ L_t^{-1}, L_t x)$ is a path in V' connecting V'_x to $V'_{x'}$.

Let $\mathbb{C}^*$ be the multiplicative group of nonzero complex numbers; then $\mathbb{C}^* \times \mathbb{C}^*$ acts freely on $\mathcal{H}_{(d)} - \{0\} \times \mathbb{C}^{n+1} - \{0\}$ by $(\lambda, \mu)(f, x) \doteq (\lambda f, \mu x)$. That the action is free means that $(\lambda, \mu)(f, x) \neq (\lambda', \mu')(f, x)$ unless $(\lambda, \mu) = (\lambda', \mu')$. This action leaves V' invariant. The equivalence relation defined by this action is the one identifying (f, x) and (g, y) if there is a (λ, μ) such that $(\lambda, \mu)(f, x) = (g, y)$. The quotient of $\mathcal{H}_{(d)} - \{0\} \times \mathbb{C}^{n+1} - \{0\}$ under this relation is $\mathbb{P}(\mathcal{H}_{(d)}) \times \mathbb{P}(\mathbb{C}^{n+1})$ and the quotient of V' is V. It follows that V is connected. To see that V is a smooth subvariety of $\mathbb{P}(\mathcal{H}_{(d)}) \times \mathbb{P}(\mathbb{C}^{n+1})$ we may first intersect V' with $S_1(\mathcal{H}_{(d)}) \times S_1(\mathbb{C}^{n+1})$, the product of the unit spheres in $\mathcal{H}_{(d)}$ and $\mathbb{C}^{n+1}$. Since V' contains the product of the lines through x and y for any $(x, y) \in V'$ it follows that V' is transversal to $S_1(\mathcal{H}_{(d)}) \times S_1(\mathbb{C}^{n+1})$ and hence $V'' = V' \cap (S_1(\mathcal{H}_{(d)}) \times S_1(\mathbb{C}^{n+1}))$ is a smooth manifold. The variety V is now the quotient of V'' by a free S^1 action. Since S^1 is compact it follows that V is a smooth manifold.

Now it follows that the codimension of V equals the codimension of V', and similarly that the tangent space is as asserted. □

Remark 2 We have taken care here to use the compact group S^1 because the quotients of smooth manifolds by free noncompact group actions, even free $\mathbb{R}$-actions, are not always smooth manifolds.

From now on we assume $k = n$. In that case $\dim V = \dim \mathbb{P}(\mathcal{H}_{(d)})$.

Our analysis concentrates on the restriction to V of the projection from $\mathbb{P}(\mathcal{H}_{(d)}) \times \mathbb{P}(\mathbb{C}^{n+1})$ onto the first factor. We denote this map by π_1:

$$\begin{tikzcd}
V \arrow[r, hook] \arrow[dr, "\pi_1"'] & \mathbb{P}(\mathcal{H}_{(d)}) \times \mathbb{P}(\mathbb{C}^{n+1}) \arrow[d] \\
& \mathbb{P}(\mathcal{H}_{(d)})
\end{tikzcd}$$

The map $\pi_1 : V \to \mathbb{P}(\mathcal{H}_{(d)})$ has the property that for each $f \in \mathbb{P}(\mathcal{H}_{(d)})$, $\pi_1^{-1}(f)$ is naturally identified with the set of zeros of f in $\mathbb{P}(\mathbb{C}^{n+1})$.

10.4 The Discriminant Variety in One Variable

We now return to the case $n = 1$. Then $\mathcal{H}_{(d)} = \mathcal{H}_d$. Version 2 of the fundamental theorem of algebra may be interpreted as saying that the fiber of $\pi_1 : V \to \mathbb{P}(\mathcal{H}_d)$

has d points everywhere as long as the points are counted with multiplicity. Now let us consider the derivative of $\pi_1 : V \to \mathbb{P}(\mathcal{H}_d)$. We are interested in the *critical variety* Σ', that is, the set $\Sigma' \subseteq V$ of points where this derivative is singular. For a nonzero tangent vector (h, v) at (f, x) in V to map to zero requires that h be zero or that $Df(x)v = 0$. That is, we simultaneously have that $f(x) = 0$ and $Df(x)(v) = 0$ where $\langle x, v\rangle = 0$. The following elementary lemma helps us identify the critical set Σ' of $\pi_1 : V \to P_{\mathcal{H}_d}$.

Lemma 2 *The pair $(f, x) \in \mathbb{P}(\mathcal{H}_d) \times \mathbb{P}(\mathbb{C}^2)$ belongs to Σ' if and only if $(\partial f/\partial z)(x) = (\partial f/\partial w)(x) = 0$ where $f(z, w)$ is an homogeneous polynomial and $x \in \mathbb{C}^2 - \{0\}$.*

Proof. If $(f, x) \in \Sigma'$ we have seen that there exists $v \in \mathbb{C}^2 - \{0\}$ such that $\langle x, v\rangle = 0$ and $Df(x)v = 0$, but also $f(\lambda x) = 0$ for $\lambda \in \mathbb{C} - \{0\}$ so $Df(x)(x) = 0$. Thus $Df(x)$ is zero at x and $(\partial f/\partial z)(x) = (\partial f/\partial w)(x) = 0$. On the other hand, if $(\partial f/\partial z)(x) = (\partial f/\partial w)(x) = 0$, then $Df(x)$ is zero; so $Df(x)(x) = 0$ and $Df(\lambda x)(x) = \lambda^{d-1} Df(x)(x) = 0$ for all λ. So f is constant on the line through x and hence x is a zero of f; so $(f, x) \in V$. Alternatively, we can see that $f(x) = 0$ by Euler's identity

$$z\frac{\partial f}{\partial z}(x) + w\frac{\partial f}{\partial w}(x), = df(x)$$

where $x = (z, w)$. □

It follows that Σ' is a subvariety of $\mathbb{P}(\mathcal{H}_d) \times \mathbb{P}(\mathbb{C}^2)$ of complex codimension 2 (it is defined by two independent equations), and hence Σ' is a subvariety of V of complex codimension 1.

Definition 4 The image $\pi_1(\Sigma') = \Sigma$ is called the *discriminant variety* $\Sigma \subset \mathbb{P}(\mathcal{H}_d)$. In other words, $f \in \Sigma$ if and only if f has a multiple root.

It is not difficult to give equations for the variety Σ itself.

Definition 5 Given two homogeneous polynomials f, g of degrees d and ℓ,

$$f(z, w) = \sum_{i=0}^{d} a_i z^i w^{d-i} \quad \text{and} \quad g(z, w) = \sum_{i=0}^{\ell} b_i z^i w^{\ell-i},$$

their *resultant* is the polynomial in their coefficients given by the determinant of the following $(\ell + d) \times (\ell + d)$ matrix (called the *Sylvester matrix*),

$$\begin{array}{l}
\ell \text{ rows} \left\{ \begin{array}{l} \\ \\ \\ \end{array} \right. \\
d \text{ rows} \left\{ \begin{array}{l} \\ \\ \\ \end{array} \right.
\end{array}
\begin{pmatrix}
a_d & \dots & a_{d-\ell+1} & \dots & \dots & a_0 & \dots & \\
 & \ddots & & & & & & \\
 & & a_d & & & \dots & a_1 & a_0 \\
b_\ell & \dots & b_0 & \dots & & & & \\
 & \ddots & & & & & & \\
 & & \ddots & & & & & \\
 & & & & b_\ell & \dots & b_1 & b_0
\end{pmatrix}$$

More precisely the entry c_{ij} in row i and column j in this matrix is given by

$$c_{ij} = \begin{cases} a_{d-j+i} & \text{for } 1 \le i \le \ell \quad \text{and} \quad i \le j \le d+i \\ b_{i-j} & \text{for } \ell+1 \le i \le \ell+d \quad \text{and} \quad i-\ell \le j \le i \\ 0 & \text{otherwise.} \end{cases}$$

The *discriminant* of an homogeneous polynomial $f(z, w)$ of degree d is the homogeneous polynomial of degree $2d-1$ in the coefficients of f defined by the resultant of $(\partial f/\partial z)$ and $(\partial f/\partial w)$.

The main property of the resultant is given in the following lemma.

Lemma 3 *The homogeneous polynomials f and g have a common zero in $\mathbb{P}(\mathbb{C}^2)$ if and only if their resultant is zero.*

Proof. The pair (r_1, r_2) is a solution of $f(z, w) = 0$ iff $(r_2 z - r_1 w)$ divides $f(z, w)$ and the same holds for g. It follows that f and g of degrees d and ℓ have a common solution if and only if there are nontrivial homogeneous polynomials k and h of degrees $\ell-1$ and $d-1$, respectively, such that $fk + gh = 0$. Viewing $(k, h) \to fk + gh$ as a linear map from $\mathcal{H}_{\ell-1} \times \mathcal{H}_{d-1} \to \mathcal{H}_{\ell+d-1}$ we see that the dimension of the spaces $\mathcal{H}_{\ell-1} \times \mathcal{H}_{d-1}$ and $\mathcal{H}_{\ell+d-1}$ are equal and therefore there is a nontrivial common zero of f and g if and only if the determinant of the linear map is zero. If

$$f(z, w) = \sum_{i=0}^{d} a_i z^i w^{d-i} \quad \text{and} \quad g(z, w) = \sum_{i=0}^{\ell} b_i z^i w^{\ell-i},$$

then this determinant is the Sylvester determinant and the lemma follows. $\square$

Proposition 2 *The discriminant variety $\Sigma \subset \mathbb{P}(\mathcal{H}_d)$ is the zero set of the discriminant polynomial.*

Proof. For any $f \in \mathbb{P}(\mathcal{H}_d)$ one has that $f \in \Sigma$ if and only if the resultant of $(\partial f/\partial z)$ and $(\partial f/\partial w)$ is zero. $\square$

Thus Σ is a subvariety of $\mathbb{P}(\mathcal{H}_d)$ and Σ also has complex codimension 1. Now it is easy to see that Σ cannot separate $\mathbb{P}(\mathcal{H}_d)$, or even any ball $B \subset \mathbb{P}(\mathcal{H}_d)$, into two connected components. For if distinct f, g lie in $B - (\Sigma \cap B)$, let L be the complex line $L = \{sf + tg \mid s, t \in \mathbb{C}\}$ considered in $\mathbb{P}(\mathcal{H}_d)$. Then $L \cap B$ is a disc in the Riemann sphere and $(L \cap B \cap \Sigma)$ has at most $2d-1$ points. So we may connect f to g in $L \cap B$ by an arc that avoids Σ. Similarly, $\mathbb{P}(\mathcal{H}_d) - \Sigma$ is connected.

Proof of Theorem 1. First we note that if $\zeta = (\zeta_1, \zeta_2)$ is a solution of $f(z, w) = 0$, then $\zeta_2 z - \zeta_1 w$ divides $f(z, w)$. So $f(z, w)$ has at most d solutions. Now let $r_1, \ldots, r_d \in \mathbb{C}$ such that $r_i \ne r_j$ for $i \ne j$. Let $g(z, w) = (z - r_1 w) \cdots (z - r_d w)$. Then g has d solutions. We consider f and g in $\mathbb{P}(\mathcal{H}_d)$. Then $g \notin \Sigma$. If h is any other element of $\mathbb{P}(\mathcal{H}_d) - \Sigma$, h and g may be joined by an arc g_t, $0 \le t \le 1$, in $\mathbb{P}(\mathcal{H}_d) - \Sigma$ such that $g_0 = g$ and $g_1 = h$. By the inverse function theorem applied to the map $\pi_1 : V - \pi_1^{-1}(\Sigma) \to \mathbb{P}(\mathcal{H}_d) - \Sigma$ we may lift the arc g_t to $(g_t, r_{it}) \subset V$ starting at $(g_0, (r_i, 1))$ for each i. Each of these d distinct arcs gives a solution of h for $t = 1$. Hence h has d solutions.

Thus if $f \in \mathbb{P}(\mathcal{H}_d) - \Sigma$, we are done. If otherwise $f \in \Sigma$, there is a sequence $g_i \in \mathbb{P}(\mathcal{H}_d) - \Sigma$, $i = 1, 2, \ldots$ such that g_i converges to f. Let ζ_i be a solution of g_i. Then since V is compact, the sequence (g_i, ζ_i) has a limit point (f, ζ) in V and $f(\zeta) = 0$. Thus f has at least one solution.

Now we show that the sum of the multiplicities of the roots of f is d. To do this we take a different definition of multiplicity, one which will go over to the proof of Bézout's Theorem. Let $\zeta_1, \ldots, \zeta_k$ be the roots of f and let U_i be disjoint open neighborhoods of the ζ_i, $i = 1, \ldots, k$. Since f is not zero in the compact set $\mathbb{P}(\mathbb{C}^2) - \bigcup_{i=1}^k U_i$, by continuity there is a ball B around f in $\mathbb{P}(\mathcal{H}_d)$ such that any $g \in B$ has all its roots in $\cup_{i=1}^k U_i$. Let $g \in B - \Sigma$ and let the multiplicity of ζ_i be the number of roots of g in U_i. The sum of the multiplicities is d since g has d roots. We now show that this definition makes sense. If g_0 and g_1 are both in $B - \Sigma$, they may be connected by an arc g_t in $B - \Sigma$ since Σ does not separate B. Now we may lift the arc g_t to d arcs in V as for the preceding homotopy (we already know that each g_t has d roots, since g_t is not in Σ). Since all the roots of g_t stay in $\cup_{i=1}^k U_i$, g_0 and g_1 have the same number of roots in each U_i. We leave the equivalence of the definitions of multiplicity as an exercise. $\square$

We conclude with one last point, that the multiplicity of any root is at least one in our definition. Given $f, \zeta_i, i = \{1, \ldots, k\}$ as in the preceding, consider $(B \times U_i) \cap V$. This is an open set that contains (f, ζ_i), and contains points not in Σ'. Thus $\pi_1((B \times U_i) \cap V)$ contains an open set in $\mathbb{P}(\mathcal{H}_d)$ and $\pi_1((B \times U_i) \cap V)$ contains points $g \in B - \Sigma$. Such a g has a root in U_i.

Remark 3 From the inverse function theorem it follows that the roots of a polynomial vary differentiably in the neighborhood of a polynomial that is not in the discriminant variety. From the properties of the multiplicity index it follows that the set of unordered roots of a polynomial written with multiplicity vary continuously. We formalize these statements in the next corollary. Our roots lie in $\mathbb{P}(\mathbb{C}^2)$. So d unordered roots lie in $\mathbb{P}(\mathbb{C}^2)^d/S_d$ where S_d is the symmetric group that acts on $\mathbb{P}(\mathbb{C}^2)^d$ by permuting the factors. Our polynomial systems lie in $\mathbb{P}(\mathcal{H}_d)$. The topological space $\mathbb{P}(\mathbb{C}^2)^d/S_d$ is compact and is a differentiable manifold in the complement of the large diagonal Δ. As a consequence of Theorem 1 we obtain a differentiable version of the classical continuity of the roots as a function of the coefficients.

Corollary 1 *The function*

$$\sigma : \mathbb{P}(\mathcal{H}_d) \longrightarrow \mathbb{P}(\mathbb{C}^2)^d/S_d$$

which takes polynomial equations to their unordered roots is continuous in $\mathbb{P}(\mathcal{H}_d)$ and differentiable on $\mathbb{P}(\mathcal{H}_d) - \Sigma$. $\square$

10.5 Bézout's Theorem

Bézout's Theorem is the n-dimensional version of the Fundamental Theorem of Algebra. It deals with n homogeneous polynomial equations in $n+1$ unknowns. The relationship with n polynomial equations in n unknowns is the same that we have already seen setting one of the homogeneous variables equal to 1.

Even for two homogeneous linear equations in three unknowns we may have a whole line of solution lines without the system of equations being identically zero. In this case the solutions of nearby problems cannot vary uniformly continuously with the problem. Arbitrarily close to a problem with a whole line of solutions are problems with unique solutions that are more than $\pi/4$ apart. Since projective space is compact and continuous functions on compact spaces are uniformly continuous, the roots of polynomial systems cannot vary continuously. In general the solutions of a system of n homogeneous polynomial equations $f = (f_1, f_2, \ldots, f_n)$ in $n+1$ unknowns may not be finite and may have quite complicated geometry. We denote by $\mathcal{D}$ the *Bézout number* of f given by

$$\mathcal{D} = \prod_{i=1}^{n} d_i,$$

where d_i is the degree of f_i for $i = 1, \ldots, n$.

The definitions of Σ and Σ' of the previous section may be extended to the n-dimensional case as follows.

The subvariety $\Sigma' \subset V$ is defined as the set of critical points of the projection $\pi_1 : V \to \mathbb{P}(\mathcal{H}_{(d)})$ and $\Sigma = \pi_1(\Sigma')$. From Proposition 1 it follows that a point $(f, x) \in \Sigma'$ if and only if $f(x) = 0$ and there is a $w \in T_x$, $w \neq 0$ such that $Df(x)w = 0$; that is, x is a degenerate zero of f. Thus $f \in \Sigma$ if and only if f has a degenerate zero. Although it is not immediately obvious that the conditions $f(x) = 0$, $Df(x)w = 0$, $w \in T_x$, and $w \neq 0$ define an algebraic variety we may rephrase them as $f(x) = 0$ and rank $Df(x) < n$ or yet as $f(x) = 0$ and $\det(M) = 0$ for all $n \times n$ minors M of $Df(x)$. This last is a system of polynomial equations and we see that Σ' is an algebraic subvariety of the smooth connected variety V. The image $\Sigma \subset \mathbb{P}(\mathcal{H}_{(d)})$ is likewise an algebraic subvariety, the discriminant variety. (We prove that Σ is a variety in Theorem 2 of Appendix B.)

Note that the polynomial system $f_0 = (f_{01}, \ldots, f_{0n})$, where

$$f_{0i}(z_1, \ldots, z_{n+1}) = z_i^{d_i} - z_{n+1}^{d_i}$$

has $\mathcal{D}$ roots and they are all nondegenerate. Thus $f_0 \notin \Sigma$ and $\Sigma \neq \mathbb{P}(\mathcal{H}_{(d)})$.

We now state and prove Bézout's Theorem in two versions, an extended geometric version and a more elementary version.

Theorem 2 (Bézout's Theorem) *Suppose $f \in \mathcal{H}_{(d)} - \Sigma$. Then f has $\mathcal{D}$ zeros in $\mathbb{P}(\mathbb{C}^{n+1})$.*

Proof. It follows just as in the previous section that $\mathbb{P}(\mathcal{H}_{(d)}) - \Sigma$ is connected. Now by the inverse function theorem it follows that any two systems f_1 and $f_2 \in \mathbb{P}(\mathcal{H}_{(d)}) - \Sigma$ have the same number of zeros since they may be connected by an arc in $\mathbb{P}(\mathcal{H}_{(d)}) - \Sigma$, and this number is $\mathcal{D}$ from the preceding example. This finishes the proof of Theorem 2. □

The next theorem extends Bézout's Theorem to the case of multiple or even infinitely many zeros. The index m is the multiplicity.

Theorem 3 (Bézout's Theorem—Extended Geometric Version) *Let $f_i : \mathbb{C}^{n+1} \to \mathbb{C}$, $i = 1, \ldots, n$ be homogeneous polynomials of degree d_i, not all identically zero, and let $f = (f_1, \ldots, f_n)$. Let $Z_j = Z_j(f)$, $j = 1, \ldots, k$ be the connected components of the set of zeros of f in $\mathbb{P}(\mathbb{C}^{n+1})$. Then we may assign an index $m(Z_j)$ to each Z_j that satisfies*

(a) *$m(Z_j) \geq 1$ for $j = 1, \ldots, k$;*

(b) *$\sum_{j=1}^{k} m(Z_j) = \mathcal{D}$;*

(c) *$m(Z_j) = 1$ for a nondegenerate isolated zero; and*

(d) *there are neighborhoods U_j of Z_j in $\mathbb{P}(\mathbb{C}^{n+1})$, for $j = 1, \ldots, k$, and B of f in $\mathbb{P}(\mathcal{H}_{(d)})$ such that if $g = (g_1, \ldots, g_n) \in B$, then $Z(g) \subset \cup_{j=1}^{k} U_j$ and $\sum_t m(Z_t(g)) = m(Z_j)$ where the sum is taken over all components of $Z(g)$ contained in U_j.*

Proof. If $f \in \Sigma$, let $f_i \in \mathbb{P}(\mathcal{H}_{(d)}) - \Sigma$ be a sequence such that f_i converges to f. Let $(f_i, \zeta_i) \in V$ and (f, ζ) a limit point of the sequence (f_i, ζ_i). Then $f(\zeta) = 0$. So $Z(f)$ is not empty.

Now let $Z_j = Z_j(f)$, $j = 1, \ldots, k$ be the connected components of the zeros of $f \in \mathbb{P}(\mathcal{H}_{(d)})$, where f may be in Σ. Let U_j be disjoint open sets with $Z_j \subset U_j$, $j = 1, \ldots, k$. By continuity of the polynomial system f it follows that there is a ball B around f in $\mathbb{P}(\mathcal{H}_{(d)})$ such that all the zeros of any $g \in B$ are contained in $\cup_{j=1}^{k} U_j$. Suppose $g \in B - \Sigma$ and define $m(Z_j)$ to be the number of zeros of g in U_j. This is well-defined since $B - \Sigma$ is connected. Properties (b), (c), and (d) of the index m follow directly from the definition.

We now turn to the proof of (a). Let $V_j = (B \times U_j) \cap V$. Then V_j is an open set in V that contains $\{f\} \times Z_j$. Since Σ' is a proper subvariety of the connected variety V, V_j cannot be contained in Σ. Hence the projection of V_j into B contains an open set and a point $g_j \in B - \Sigma$. By definition $\pi_1^{-1}(g_j)$ has at least one point in V_j; that is, g_j has at least one solution in U_j and $m(Z_j) > 0$. □

10.6 Additional Comments and Bibliographical Remarks

Bézout's Theorem is a classical theorem in algebraic geometry. See, for example, [Mumford 1976; Shafarevich 1977]. The approach taken here follows closely

[Shub 1993b] and [Shub and Smale 1993a, 1993b, 1993c, 1996, 1994]. The resultant of two one-variable polynomials is described in [Lang 1993a] and more generally in [Van der Waerden 1949] (old edition of "modern algebra" and "algebraic geometry"). See [Griffiths and Harris 1978] for background on projective space and complex algebraic geometry. Background in differential topology can be found in [Guillemin and Pollack 1974; Hirsch 1976; Milnor 1965].

11

Condition Numbers and the Loss of Precision of Linear Equations

The condition number of an invertible real or complex $n \times n$ matrix A is defined as

$$\kappa(A) = \|A\| \, \|A^{-1}\|,$$

where $\|A\|$ is the operator norm

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|}$$

and $\mathbb{R}^n$ or $\mathbb{C}^n$ is given the usual inner product. The condition number measures the relative error in the solution of the system of linear equations

$$Ax = v.$$

If the error in v is δv causing an error δx so that $A(x + \delta x) = v + \delta v$, then

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\delta v\|}{\|v\|}. \tag{11.1}$$

Here is the argument

$$\frac{\|\delta x\|}{\|x\|} = \frac{\|\delta x\|}{\|v\|}\frac{\|v\|}{\|x\|} = \frac{\|A^{-1}\delta v\|}{\|v\|}\frac{\|Ax\|}{\|x\|} \leq \|A^{-1}\|\|A\|\frac{\|\delta v\|}{\|v\|} = \kappa(A)\frac{\|\delta v\|}{\|v\|}.$$

Moreover, it is easily seen that there are vectors v and δv such that $\|\delta x\|/\|x\| = \kappa(A)\|\delta v\|/\|v\|$, where $x = A^{-1}v$ and $\delta x = A^{-1}(v + \delta v) - x$. So $\kappa(A)$ is a sharp worst case estimate of the relative error in x, as a function of the relative error in v.